

HW05 – AI-assisted Performance Testing

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Repository: <<https://github.com/linhnph05/hw05>>

Tools and environment

Item	Value
SUT	EShop Node.js + Express + SQLite backend at `http://127.0.0.1:3000`
Load tool	Apache JMeter 5.6.3, non-GUI mode
Resource monitor	macOS `ps`, sampled once per second for the backend PID
Hardware	MacBook Pro (MacBookPro17,1), Apple M1, 8 CPU cores, 16 GB memory
OS	macOS 15.7.3
AI assistance	A second Codex/GPT-5.6 agent; full log in `AI_Audit_Report.md`

I used JMeter non-GUI mode for the measured runs. This avoids using the GUI listener as a load generator. The raw JTL files, JMeter HTML reports, JMeter logs, and resource CSV files are in `results/`.

Commands used for the measured runs

I ran these commands from the repository root. `-n` runs JMeter without the GUI, `-l` writes the raw JTL, and `-e -o` generates the HTML report.

```
```sh
jmeter -n -t test-plans/23127081_Load_20260816.jmx -l results/load.jtl -e -o results/load_html -j results/load_jmeter.log
jmeter -n -t test-plans/23127081_Stress_20260816.jmx -l results/stress.jtl -e -o results/stress_html -j results/stress_jmeter.log
jmeter -n -t test-plans/23127081_Spike_20260816.jmx -l results/spike_baseline.jtl -e -o results/spike_baseline_html -j results/spike_baseline_jmeter.log -Jspike_threads=2 -Jspike_ramp=1 -Jspike_stage_duration=60
jmeter -n -t test-plans/23127081_Spike_20260816.jmx -l results/spike.jtl -e -o results/spike_html -j results/spike_jmeter.log -Jspike_threads=30 -Jspike_ramp=5 -Jspike_stage_duration=60
jmeter -n -t test-plans/23127081_Spike_20260816.jmx -l results/spike_recovery.jtl -e -o results/spike_recovery_html -j results/spike_recovery_jmeter.log -Jspike_threads=2 -Jspike_ramp=1 -Jspike_stage_duration=60
jmeter -n -t test-plans/23127081_Load_20260816.jmx -l results/endurance.jtl -e -o results/endurance_html -j results/endurance_jmeter.log -Jload_threads=100 -Jload_ramp=60 -Jload_duration=600
```
```

I started `trash/monitor\_backend.sh` at the same time as each run. It sampled the backend process with `ps` once per second and wrote the matching resource CSV file in `results/`.

JMeter HTML report screenshots

I opened the generated JMeter HTML dashboards in Chrome and captured the visible dashboard summaries. These screenshots are copies of the rendered reports, not recreated result tables.

![Load JMeter dashboard](../images/load_jmeter_html_report.jpg)

![Stress JMeter dashboard](../images/stress_jmeter_html_report.jpg)

![Spike JMeter dashboard](../images/spike_jmeter_html_report.jpg)

![Endurance JMeter dashboard](../images/endurance_jmeter_html_report.jpg)

Endpoint scope

I selected three endpoints in each scenario.

| Scenario and plan | Endpoint group | Endpoint workflow | CSV input | Listener/report view |
|-------------------------------------|----------------|---|-----------------------------------|----------------------|
| Load - `23127081_Load_20260816` | Read-heavy | `GET /api/cart` → `GET /api/orders/my-orders` → `GET /api/orders/{id}` | `test-data/read_input.csv` | Summary Report |
| Stress - `23127081_Stress_20260816` | Auth-heavy | `POST /api/register` → `POST /api/forgot-password` → `POST /api/reset-password` | `test-data/auth_input.csv` | Aggregate Report |
| Spike - `23127081_Spike_20260816` | Transactional | `POST /api/apply-coupon` → `POST /api/admin/coupons` → `DELETE /api/admin/coupons/{id}` | `test-data/transaction_input.csv` | View Results Tree |

Each plan has status-code assertions. The stress plan extracts the reset token from the forgot-password response before calling reset-password. The spike plan extracts the temporary coupon ID and deletes the temporary coupon, so its write test cleans up after itself.

AI design and my review

I first asked the AI for endpoint and parameter ideas. It suggested one endpoint for each scenario. That would meet the HW05 endpoint-group rule, but I chose a broader scope of three endpoints in each workflow. I added the CSV files, HTTP assertions, reset-token extractor, and distinct JMeter listeners.

The AI proposed a 50-user stress run. I used 30 users for the first real stress run because this is a student laptop and the SUT writes to one SQLite database. The final parameters were based on actual smoke tests, not only on the AI answer. The complete prompts and responses are in the audit report.

Measured results

| Run | Parameters | Samples | Throughput | Mean | p95 | Maximum | Errors |
|----------------|--|---------|------------|----------|--------|---------|--------|
| Load | 20 users, 60 s ramp, 300 s duration, 1 s think time | 5,348 | 17.8 RPS | 8.56 ms | 30 ms | 178 ms | 0.00% |
| Stress | 30 users, 60 s ramp, 180 s duration, 500 ms think time | 8,522 | 47.3 RPS | 25.06 ms | 139 ms | 648 ms | 0.00% |
| Spike baseline | 2 users, 1 s ramp, 60 s duration | 230 | 3.83 RPS | 10.43 ms | 36 ms | 139 ms | 0.00% |
| Spike | 30 users, 5 s ramp, 60 s duration | 3,255 | 54.25 RPS | 23.37 ms | 103 ms | 622 ms | 0.00% |
| Spike recovery | 2 users, 1 s ramp, 60 s duration | 226 | 3.77 RPS | 14.84 ms | 40 ms | 377 ms | 0.00% |

The spike test used three time-ordered stages. I merged their raw stage JTL files into `results/spike.jtl`; `results/spike\_execution.md` records each stage. The temporary stage files are in `trash/` and are not submission files.

Resource use

| Run | Average CPU | Peak CPU | Average RSS | Peak RSS |
|--------|-------------|----------|-------------|-----------|
| Load | 1.99% | 17.3% | 33,530 KB | 45,744 KB |
| Stress | 9.17% | 24.6% | 42,382 KB | 64,304 KB |

| Spike | 3.72% | 18.0% | 34,357 KB | 63,664 KB |

`ps` monitored the actual `/opt/homebrew/bin/node server.js` backend process, not the JMeter process. This is why the backend CPU values are modest even when the test generator uses many threads.

Endurance threshold

I ran the read workflow for 10 minutes with 100 users, a 60-second ramp, and a one-second think time. The run completed 55,622 samples at **92.70 RPS**, with mean 19.04 ms, p95 83 ms, maximum 1,031 ms, and 0.00% errors. Backend peak CPU was 26.9% and peak RSS was 76,000 KB.

Therefore, the measured stable level on this machine is **at least 92.70 RPS for ten minutes at 100 users**. It is not an absolute hardware maximum: the backend still had CPU headroom and I did not run the machine until it failed.

AI analysis and misinterpretation hunt

The second AI calculated useful results and proposed performance gates. I checked the raw JTL values myself. One small metric error was found: it reported the merged spike p95 as 97 ms and the spike-stage p95 as 105 ms. Sorting the raw elapsed values in `results/spike.jtl` gives a merged p95 of **93 ms**; the spike-stage raw file gives **103 ms**. The error likely came from a percentile index or rounding choice. I use my direct raw-JTL values in this report.

The AI correctly warned that 92.89 RPS must not be called the hardware limit. My own calculation is 92.70 RPS because I divide the 55,622 samples by the planned 600-second run duration. These are different denominator choices, not evidence of a regression.

| AI recommendation | My judgement | Reason |
|------------------------------------|---------------------------|--|
| --- --- --- | | |
| Add index on `orders(user_id, id)` | Feasible | The order-history query filter s by `user_id` and sorts by ID. Measure query plans before changing it. |
| Add index on `users(email)` | Feasible | Register, forgot-password, and login use email lookup. |
| Enable SQLite WAL | Feasible experiment | It may help concurrent writes, but these runs do not prove a need. Measure before and after. |
| Add a normal connection pool | Hallucinated / unsuitable | This SUT uses one local SQLite connection, not a remote database service. |
| Add caching now | Not justified | CPU is low and there were no errors. First identify a slow database query. |

Continuous performance testing proposal

```
graph TD
    A[Commit or pull request] --> B{Backend, database, or API path changed?}
    B -- No --> C[Skip full performance suite]
    B -- Yes --> D[Build isolated EShop service]
    D --> E[Run smoke JMeter plan]
    E --> F{Smoke errors?}
    F -- Yes --> G[Fail pipeline and attach JTL]
    F -- No --> H[Run short baseline load test]
    H --> I[Compare p95, error rate, and RPS with baseline]
    I --> J{p95 +20%, errors >1%, or RPS -10%?}
    J -- Yes --> K[Flag regression and save HTML report]
    J -- No --> L[Publish result and update baseline after review]
```

Run the smoke test on every relevant pull request. Run the longer stress and

endurance suites nightly or before release, because they cost more time and compute. A p95 comparison can find a slow change early, but it can also cause false alarms when a shared CI runner is busy. To reduce false alarms, run the same plan twice before failing a release and keep the test data and hardware class fixed. This model costs CI minutes and test-database resets, but it gives the team a performance history instead of discovering regressions after merge.

AI critique

See `AI\_Critique.md` for the required 200–300 word critique.

Video note

No video was recorded, as instructed. `Video\_Narration\_and\_Steps.md` gives a Vietnamese narration script and exact recording steps for a later recording.